CardioSurv

tasks_3 — Report & Slides Assignments

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Member	Report Section	Slides	Priority
Bougacha Mohamed	Abstract (a) + Introduction (c) + Conclusion (g)	Slide 7	**HIGH**
Ang Jing Ru	Literature Review §3 (co-owner)	Slides 3, 4	**HIGH**
Law Zi Ying	Literature Review §3 (co-owner)	Slide 8	**HIGH**
Chiluba Chifunilo	Methodology §4	Slides 5, 6	**HIGH**
Tan Guan Han	Results & Discussion §5	Slides 9, 10	**HIGH**

Read before starting: `CardioSurv_Report_Briefing.md` and `CardioSurv_Report.tex/.pdf` are the single source of truth for every number, figure, and result. **Do not invent statistics.** Do not write another member's section.

Citation rule: Every claim backed by external literature must be cited using the numbered format [**1**], [**2**], [**3**]... at the point of the claim — not grouped at the end of a paragraph. Aim for **at least 3–5 citations per sub-section** and a minimum of **15 unique references** across the full report. Use Google Scholar, IEEE Xplore, ScienceDirect, SpringerLink, PubMed. No Wikipedia.

1 Section 1 — Report Assignments

1.1 Bougacha Mohamed — Abstract + Introduction + Conclusion

BOUGACHA MOHAMED Abstract (a) • Introduction (c) • Conclusion (g) **HIGH PRIORITY**

Role. You write the three framing sections of the report: the Abstract (written last), the Introduction (written first, sets the clinical motivation), and the Conclusion (written after the Results are done). These are the sections the examiner reads most carefully. Do not start the Abstract until every other section is complete.

Task B-1 Abstract HIGH PRIORITY Write LAST — after all sections done

What to write. One single paragraph. **250–300 words. No citations. No bullet points.** Must flow as continuous prose and cover all five elements in this order:

1. **Objective:** End-to-end ML pipeline for cardiovascular risk triage (classification → treatment recommendation → survival prediction).
2. **Dataset:** Merged Kaggle Heart Failure dataset + UCI Cleveland Heart Disease dataset (1,221 rows after harmonisation); three engineered clinical features (AgeBin, BP_RiskLevel, HeartRateStressIndex).
3. **Methods:** Random Forest vs. XGBoost (Part 1 classifier); XGBoost recommender + Kaplan–Meier + Cox Proportional Hazards (Part 2); clinical routing via GRACE Risk Score and BED formula.
4. **Key results:** Random Forest wins all metrics — Accuracy 0.9614, F1-macro 0.9321, AUC-OVR 0.9880. Cox PH C-index 0.85. Survival data is synthetically simulated (proof-of-concept).
5. **Conclusion:** Complete two-stage clinical decision support pipeline deployed as a live web application with FastAPI backend.

Do NOT:

- Cite any references inside the Abstract.
- Use headings, bullets, or sub-sections.
- Copy sentences from the Introduction.

Target: 250–300 words, single paragraph.
Depends on: All other sections fully drafted.

Task B-2 Introduction HIGH PRIORITY Start immediately

What to write — four sub-parts in order:

(1) Background & Motivation (~150 words)

Open with the global burden of cardiovascular disease: CVD causes approximately 17.9

million deaths per year, representing 32% of all global deaths [**1**]. Narrow immediately to the clinical gap: elderly patients who appear generally healthy can still carry serious hidden cardiovascular risk due to the natural ageing of cardiac tissue and vasculature, making sudden fatal events — cardiogenic shock, acute heart failure — difficult to anticipate without systematic screening. Cite Bharati & Lev (1998) [**2**] which identifies the 50–70 year age bracket as the peak distribution for sudden cardiac death. Close with the clinical motivation: automated risk stratification from routine, non-invasive vital sign measurements has significant potential to close this gap at scale.

Figures to include: Include the EDA age distribution figure (`reports/figures/eda/age_distribution.png`) as Figure 1 with caption: "Age distribution of the study cohort. The 50–70 bracket aligns with the peak sudden cardiac death range reported in the literature [**2**]."

(2) Machine Learning in Healthcare (~80 words)

One bridging paragraph. Mention that ensemble methods — Random Forest [**3**] and XGBoost [**4**] — have demonstrated strong performance on structured clinical tabular data. Note that large EHR databases such as MIMIC-IV [**5**] have enabled richer longitudinal modelling beyond binary diagnosis. Cite at least one systematic review of ML in cardiovascular prediction [**6**]. This paragraph frames CardioSurv as grounded in established research.

(3) Project Scope (~100 words)

Three clearly numbered objectives:

1. Classify cardiovascular risk as Low, Medium, or High from 14 clinical input features, outputting a risk category and a confidence score.
2. Recommend a targeted treatment intervention (SBRT or Medication) with calibrated intensity, validated using the GRACE Risk Score [**7**] and the Biologically Effective Dose formula [**8**].
3. Estimate the patient's two-year survival probability with and without the recommended intervention, producing a counterfactual output.

State that the system is deployed as a live web application (FastAPI backend, PostgreSQL database, HTML/CSS/JS frontend hosted on Render).

(4) Selected Scenario (~30 words)

One sentence: "This project corresponds to Option A (Medical Data Analysis). The team operates as a machine learning unit within a healthcare organisation, analysing patient clinical data to support diagnosis and triage decisions."

Do NOT:

- Describe datasets or hyperparameters — those belong in Methodology.
- Present any results — those belong in Results & Discussion.
- Copy text from the assignment description sheet.

Citations to use: [**1**] WHO (2021) [**2**] Bharati & Lev (1998) [**3**] Breiman (2001) [**4**] Chen & Guestrin (2016) [**5**] Johnson et al. (2023) [**6**] Mohan et al. (2019) [**7**] Eagle et al. (2004) [**8**] Joiner & van der Kogel (2009).

Target: ~400 words + 1 figure.

Task B-3 Conclusion HIGH PRIORITY After Results section is complete

What to write — three clearly separated sub-parts:

(1) Key Findings (~120 words)

Write as a short numbered list or prose. Four findings only:

1. Random Forest outperformed XGBoost on all three metrics at this dataset scale — F1-macro 0.9321 vs. 0.9251 [**6**], confirming better recall on the minority Medium class which is the clinically critical outcome.
2. The engineered HeartRateStressIndex ranked 3rd in feature importance, validating that normalising MaxHR by age-predicted maximum captures ischaemic stress non-invasively [**9**].
3. The two-stage pipeline successfully covers all four clinical routing branches: Low, Medium, High without arrhythmia, and High with refractory ventricular tachycardia (SBRT path).
4. The Cox PH survival model achieved a concordance C-index of 0.85 [**10**], indicating strong discriminative ability within the proof-of-concept setting.

(2) Limitations (~100 words)

Four points only — do not add more:

1. The survival component uses synthetically simulated data due to MIMIC-IV access constraints [**5**] — results are proof-of-concept only and not clinically validated.
2. The merged dataset (~1,400 rows) is small for deep learning architectures.
3. No external validation on an independent site cohort was performed — generalisation to different populations is unknown.
4. The XGBoost recommender achieves perfect accuracy because intervention labels are deterministically derived from routing rules; real deployment would use noisier clinical labels.

(3) Future Work (~80 words)

Four suggestions only:

1. Replace synthetic survival data with real MIMIC-IV longitudinal records [**5**].
2. Hyperparameter optimisation via Optuna or Bayesian search for both models.
3. Add SHAP (SHapley Additive exPlanations) values [**11**] for clinical interpretability — critical for trust in medical AI deployment.
4. Design a prospective pilot study to validate risk categories against cardiologist assessment on an independent patient cohort.

Do NOT:

- Repeat the Abstract word-for-word.
- Introduce new results or concepts not discussed earlier in the report.

Depends on: Tan Guan Han's Results section must be finished first.

Target: ~300 words.

1.2 Ang Jing Ru + Law Zi Ying — Literature Review §3

ANG JING RU + LAW ZI YING Literature Review (§d) — split below **HIGH PRIORITY**

Role. You co-own the entire Literature Review. This is the most citation-intensive section in the report. You must cite **a minimum of 10 peer-reviewed sources** across the four sub-sections, with **3–5 citations per sub-section**. Every citation uses the numbered format **[N]** placed immediately after the claim it supports — not at the end of the paragraph.

ANG JING RU writes §3.1 and §3.2.
LAW ZI YING writes §3.3 and §3.4.
One person reviews the compiled section for consistency before the final PDF.

> **Source requirements:** Use only peer-reviewed academic sources: IEEE Xplore, ScienceDirect, SpringerLink, PubMed, ACM Digital Library, JAMA Network, NEJM. Every paper you cite must be directly relevant to the point you are making. Do not pad the reference list with unread papers. **Minimum 10 unique citations for the Literature Review section alone.**

Task LR-1 §3.1 ML for Cardiovascular Risk Prediction [ANG JING RU] HIGH PRIORITY Start immediately

What to write — 5 paragraphs, ≥5 citations:

Paragraph 1 — Historical context. Provide context: ML was first applied to the UCI Cleveland Heart Disease dataset in the late 1980s using logistic regression and decision trees **[12]**. Explain why structured clinical tabular data (vital signs, ECG readings, lab values) is well-suited to tree-based ensemble methods: feature interactions are non-linear, sample sizes are moderate, and interpretability matters to clinicians. Cite at least one early clinical ML study **[12]**.

Paragraph 2 — Random Forest for clinical prediction. Describe Random Forest as an ensemble of decision trees using bagging and random feature subsets **[3]**. Cite Mohan et al. (2019) **[6]**: their systematic comparison found that a hybrid Random Forest approach with feature selection achieved classification accuracy above 88% on the UCI Heart Disease dataset, outperforming Naïve Bayes, decision trees, and SVM on the same task. Mention that the `class_weight='balanced'` parameter makes RF particularly suited to imbalanced clinical datasets.

Paragraph 3 — XGBoost in healthcare. Introduce XGBoost (Chen & Guestrin, 2016 **[4]**) and its improvements over standard gradient boosting: second-order gradient statistics, L1/L2 regularisation, parallel tree construction, and native handling of missing values. Cite Christodoulou et al. (2019) **[13]**: a systematic review across 71 clinical prediction studies finding that XGBoost and Random Forest consistently outperform logistic regression on non-linear tabular data. Cite one additional clinical study using XGBoost for cardiovascular prediction **[14]**.

Paragraph 4 — Class imbalance in medical datasets. Discuss the class imbalance problem: minority clinical classes (e.g. a rare risk tier) are clinically critical but underrepresented, causing standard accuracy metrics to hide poor minority-class recall [**6**]. Explain why macro-averaged F1-score is preferred as the selection metric for imbalanced multi-class medical tasks, as it weights all classes equally regardless of frequency. Mention class-weight balancing as the technique used in CardioSurv and cite its theoretical basis [**3**].

Paragraph 5 — Feature engineering in clinical ML. Discuss how domain-informed engineered features improve model performance beyond raw measurements. Cite at least one study demonstrating the value of derived clinical features (e.g. age-normalised indices, risk-score bins) in cardiovascular ML [**9**]. This directly justifies the HeartRateStressIndex and AgeBin features engineered in CardioSurv, and strengthens the claim that the feature importance result (HRSI ranking 3rd) is clinically meaningful.

Figures to include: Include the class balance bar chart from the EDA notebook (`reports/figures/eda/class_balance.png`) as Figure N with caption: "Class distribution of the RiskCategory target variable. The Medium class represents only 8.4% of samples, motivating the use of class-weight balancing and macro-averaged F1 as the primary evaluation metric."

Do NOT:

- Present CardioSurv's own results here.
- Describe the GRACE score or BED formula — those are §3.2.
- Describe survival analysis methods — those are §3.3.

Suggested citations: [**3**] Breiman (2001) Machine Learning [**4**] Chen & Guestrin (2016) KDD [**6**] Mohan et al. (2019) IEEE Access [**9**] Shah et al. (2020) JACC Cardiovascular Imaging [**12**] Detrano et al. (1989) Am. J. Cardiology [**13**] Christodoulou et al. (2019) J. Clin. Epidemiology [**14**] Oliveira et al. (2023) Comput. Biol. Med.

Target: 5 paragraphs, ~380 words, ≥5 citations.

Task LR-2 §3.2 Clinical Scoring Systems & ML Integration [ANG JING RU]

HIGH PRIORITY Start immediately

What to write — 4 paragraphs, ≥4 citations:

Paragraph 1 — GRACE Risk Score. Describe the GRACE (Global Registry of Acute Coronary Events) score, validated by Eagle et al. (2004) [**7**] on over 11,000 ACS patients across 14 countries. The score combines age, heart rate, systolic BP, creatinine, Killip class, cardiac arrest, ST-segment deviation, and elevated enzymes. It achieves a concordance index of approximately 0.84, making it one of the strongest validated clinical risk tools in cardiology. Explain how CardioSurv uses it: patients on the medication path have their GRACE score computed to calibrate drug intensity (Standard / Intensified / Maximal). Cite the original GRACE paper [**7**] and one subsequent validation study [**15**].

Paragraph 2 — BED formula and cardiac radioablation. Introduce the Biologically Effective Dose formula from Joiner & van der Kogel (2009) [**8**]: BED $= D(1 + d/(\alpha/\beta))$, where D is total dose, d is dose per fraction, and $\alpha/\beta = 10\,\mathrm{Gy}$ for cardiac tissue. Cite Cuculich et al. (2017) [**16**]: the landmark NEJM clinical trial demonstrating that non-invasive cardiac SBRT at 25 Gy in a single fraction (BED = 87.5 Gy) suppresses refractory ventricular tachycardia in patients where catheter ablation and antiarrhythmic drugs had failed. Explain that CardioSurv uses this exact dose window as the SBRT validation step.

Paragraph 3 — Hybrid clinical-ML systems. Argue that embedding validated scoring systems directly into ML pipelines — rather than treating all features as black-box inputs — anchors decision boundaries to known clinical relationships and improves interpretability [**17**]. Cite one review of clinical decision support systems showing that hybrid (rule-based + ML) approaches earn higher clinician trust than pure data-driven models [**17**].

Paragraph 4 — Precedent for GRACE in ML. Cite at least one published study that integrates the GRACE score as a feature in an ML model for cardiovascular outcome prediction [**15**]. This strengthens the argument that CardioSurv's approach is consistent with established clinical ML practice.

Do NOT:

- Present CardioSurv's Part 2 results.
- Discuss survival analysis here — that is §3.3.

Suggested citations: [**7**] Eagle et al. (2004) JAMA [**8**] Joiner & van der Kogel (2009) Basic Clinical Radiobiology [**15**] Fox et al. (2006) BMJ (GRACE validation) [**16**] Cuculich et al. (2017) NEJM [**17**] Shortliffe & Sepulveda (2018) JAMA

Target: 4 paragraphs, ~300 words, ≥4 citations.

Task LR-3 §3.3 Survival Analysis in Clinical ML [LAW ZI YING] HIGH PRIORITY

Start immediately

What to write — 4 paragraphs, ≥4 citations:

Paragraph 1 — Survival analysis as a task. Explain that survival analysis differs from standard classification: the outcome variable is *time to event*, and observations may be censored (patient leaves the study before the event occurs). Introduce the two key challenges: censoring and the need for probabilistic outputs over time, not just a binary label [**18**]. Cite Wang et al. (2019) [**18**] as the main survey reference for this sub-section.

Paragraph 2 — Kaplan–Meier estimator. Describe the Kaplan–Meier estimator (Kaplan & Meier, 1958 [**19**]) as the standard non-parametric method for survival curve estimation from censored data. It makes no distributional assumptions and produces group-stratified survival probability curves over time. The log-rank test tests whether two survival curves differ significantly [**19**]. In CardioSurv, KM curves are stratified by treatment branch (SBRT vs. Medication).

Paragraph 3 — Cox Proportional Hazards model. Introduce Cox PH (Cox, 1972 [**10**]) as a semi-parametric regression modelling the hazard as a baseline hazard multiplied by an exponential linear combination of covariates. Its key advantage: produces per-patient survival probability estimates at any time horizon, not just group curves — essential for CardioSurv's counterfactual output. The concordance index (C-index) summarises discriminative ability: $C = 0.5$ is random chance; $C \geq 0.7$ is clinically useful [10]. CardioSurv achieves $C = 0.85$.

Paragraph 4 — Deep learning vs. classical at small scales. Cite Wang et al. (2019) [18]: deep survival models (DeepSurv, survival forests, neural Cox) outperform classical Cox PH on large high-dimensional datasets, but Cox PH remains competitive and substantially more interpretable on small-to-medium clinical tabular datasets. Cite Katzman et al. (2018) [20] (DeepSurv paper) to acknowledge the deep learning alternatives exist. Conclude: Cox PH is the appropriate choice for CardioSurv given its dataset scale (~1,400 rows) and clinical interpretability requirement.

Figures to include: Reference the Kaplan–Meier curves figure (`reports/figures/km_curves.png`) as Figure N with caption: "Kaplan–Meier survival curves stratified by treatment branch (SBRT vs. Medication) with 95% confidence bands. The log-rank test confirms significantly different survival trajectories between the two branches ($p < 0.05$)."

Do NOT:

- Present the C-index = 0.85 result here as your result — just define what C-index means. The result is presented in Results & Discussion (§5).
- Discuss MIMIC-IV here — that is §3.4.

Suggested citations: [10] Cox (1972) JRSS-B [18] Wang et al. (2019) ACM Computing Surveys [19] Kaplan & Meier (1958) JASA [20] Katzman et al. (2018) BMC Medical Informatics

Target: 4 paragraphs, ~270 words, ≥4 citations.

Task LR-4 §3.4 EHR Databases and MIMIC-IV [LAW ZI YING] MEDIUM PRIORITY

Start immediately

What to write — 2 paragraphs, ≥3 citations:

Paragraph 1 — What MIMIC-IV is and why it matters. The MIMIC-IV database (Johnson et al., 2023 [5]) is the largest freely accessible de-identified critical care EHR database, comprising over 40,000 ICU stays from Beth Israel Deaconess Medical Center. It contains structured data including vital signs, laboratory results, medication records, ICD-9/ICD-10 diagnosis codes, and discharge summaries. Access requires CITI training certification. MIMIC-IV provides exactly the longitudinal follow-up data (treatment history, survival times, medication records) that the Kaggle and UCI datasets lack. Cite one published ML study that used MIMIC-IV for cardiovascular prediction [21] to establish that this approach has academic precedent.

Paragraph 2 — Synthetic data as a valid approach. Because CITI credentialing introduced a timeline constraint during this project's development period, survival time durations and event labels were synthetically simulated using clinically grounded rules (GRACE score thresholds and arrhythmia prevalence rates from the literature). Cite Burton et al. (2006) [22]: simulation-based survival data generation for pipeline prototyping is an accepted practice in clinical ML research when real data access is restricted. Cite one additional study that explicitly uses simulated clinical data for model validation [**22**]. Frame this transparently: the survival component is a proof-of-concept demonstration.

Suggested citations: [**5**] Johnson et al. (2023) Scientific Data [**21**] Purushotham et al. (2018) J. Biomed. Informatics [**22**] Burton et al. (2006) Statistics in Medicine
Target: 2 paragraphs, ~200 words, $\geq$3 citations.

1.3 Chiluba Chifunilo — Methodology §4

CHILUBA CHIFUNILO Methodology (§e) — all sub-sections **HIGH PRIORITY**

Role. You own the entire Methodology section. This is the longest and most technically detailed section of the report. Every pipeline step must be described precisely enough that a reader could reproduce the results from your description alone. All numbers, decisions, and configurations come from `CardioSurv_Report_Briefing.md`.

Task M-1 §4.1 System Overview + Pipeline Diagram HIGH PRIORITY Start immediately

What to write. One paragraph (~80 words) introducing the two-stage pipeline, followed by the system flowchart figure.

- Describe CardioSurv as a two-stage ML pipeline connected by a rule-based clinical routing layer. Stage 1 classifies cardiovascular risk from patient vital signs. Stage 2 recommends an intervention and estimates survival probability.
- Include the system flowchart figure (already built in `CardioSurv_Report.tex` as a TikZ diagram) with caption: "End-to-end system flowchart of the CardioSurv ML pipeline. The SBRT branch (red) is triggered for High-risk patients with arrhythmia; all other patients are routed to the Medication branch (green)."
- Reference the figure: "The complete pipeline is shown in Figure N."

Target: ~80 words + pipeline figure.

Task M-2 §4.2 Datasets HIGH PRIORITY Start immediately

What to write — 3 paragraphs:

- **Paragraph 1 — Kaggle Heart Failure dataset [23]:** 918 rows, 11 features. List all features explicitly: Age, Sex, ChestPainType, RestingBP, Cholesterol, FastingBS, RestingECG, MaxHR, ExerciseAngina, Oldpeak, ST_Slope. This is the primary source for classification labels and features.
- **Paragraph 2 — UCI Cleveland dataset [12]:** 303 rows, 14 features. Adds two clinically significant angiographic predictors: `ca` (number of major vessels coloured by fluoroscopy, 0–4) and `thal` (thalassemia type: normal, fixed defect, reversible defect). Explain what fluoroscopy is in one sentence.
- **Paragraph 3 — Merge rationale + synthetic survival:** Merging after harmonising column names and encoding schemes produced 1,221 rows with a richer feature set. State clearly that the original datasets contain no longitudinal follow-up data. Because MIMIC-IV [5] access was constrained by CITI credentialing timelines, survival time durations and event labels were synthetically simulated using clinically grounded rules (GRACE score, arrhythmia prevalence rates). Frame this as a principled proof-of-concept decision.

Figures to include: Present the dataset summary as a formatted table: Dataset | Source | Rows | Key features added.
Target: ~300 words + table.

Task M-3 §4.3 Data Cleaning + Preprocessing Pipeline HIGH PRIORITY Start immediately

What to write:

- **Cleaning steps** (bullet list): missing value imputation (column medians for numeric, mode for categorical); removal of duplicate rows; type coercion for consistent dtypes across the merged set; Cholesterol = 0 treated as missing and imputed (clinically impossible value).
- **Preprocessing pipeline:** Describe the scikit-learn `ColumnTransformer` applied uniformly to both models. Numeric features (Age, RestingBP, Cholesterol, FastingBS, MaxHR, Oldpeak, HeartRateStressIndex) → `StandardScaler` (zero mean, unit variance). Categorical features (Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope, AgeBin, BP_RiskLevel) → `OneHotEncoder` with `drop='first'` to avoid the dummy variable trap. Explain in one sentence why `drop='first'` is necessary (prevents perfect multicollinearity between encoded columns).
- **Train/test split:** 80/20 stratified split on `RiskCategory` (`random_state=42`). Explain why stratification is essential: the Medium class = 8.4% of samples (119 of 1,421 rows); without stratification the test set may contain too few Medium-class samples for reliable evaluation. Final: 1,136 training rows, 285 test rows.

Target: ~200 words.

Task M-4 §4.4 Feature Engineering + EDA Summary HIGH PRIORITY Start immediately

What to write:

- **Feature engineering table** (Table N): Three rows with columns: Feature | Formula | Clinical motivation.
  - **AgeBin:** `<40 / 40-49 / 50-59 / 60-69 / 70+` — captures non-linear age effects; risk steps up sharply after 50 and 70.
  - **BP_RiskLevel:** AHA categories Normal (<120) / Elevated (120–129) / Stage 1 (130–139) / Stage 2 (140–179) / Crisis (≥180 mmHg) — converts continuous BP into clinically actionable categories aligned with treatment guidelines.
  - **HeartRateStressIndex:** HRSI = MaxHR/(220 − Age) — normalises achieved HR against age-predicted maximum; values near 1.0 indicate full chronotropic competence; blunted response (<0.6) is associated with myocardial ischaemia [9].
- **EDA summary (4–5 sentences):** State that EDA was conducted in `01_eda.ipynb` (Tan Guan Han). Key findings: class balance 55% positive / 45% negative; Medium risk

= 8.4% minority class; age distribution centred at 54 years (consistent with the 50–70 peak risk bracket [2]); ASY chest pain type is the strongest categorical predictor of heart disease; Oldpeak and MaxHR are the strongest numeric correlates.

Figures to include:

- Numeric distributions figure (`reports/figures/eda/numeric_distributions.png`).
- Spearman correlation heatmap (`reports/figures/eda/correlation_heatmap.png`).

Target: ~220 words + table + 2 figures.

Task M-5 §4.5 Part 1 Models + §4.6 Part 2 Models HIGH PRIORITY Start immediately

Part 1 — what to write:

- **Task definition:** multi-class classification, `RiskCategory` ∈ {Low, Medium, High}, 14 input features.
- **Random Forest [3]:** n_estimators=300, max_depth=None (fully grown trees), class_weight='balanced', random_state=42. One sentence: `class_weight='balanced'` reweights each sample inversely proportional to its class frequency, compensating for the 8.4% Medium class.
- **XGBoost [4]:** n_estimators=400, max_depth=6, learning_rate=0.05, objective='multi:softprob', num_class=3, eval_metric='mlogloss', random_state=42.
- Both models wrapped in a scikit-learn `Pipeline` (preprocessor + classifier) to prevent data leakage.
- **Evaluation metrics:** Accuracy, F1-macro (primary selection metric — weights all classes equally, critical for the imbalanced Medium class), AUC-OVR (macro-averaged One-vs-Rest).

Part 2 — what to write:

- **Clinical routing rule:** High-risk + arrhythmia → SBRT branch; otherwise → Medication branch. Present the BED formula as a display equation with $D = 25\,\mathrm{Gy}$, $d = 25\,\mathrm{Gy}$, $\alpha/\beta = 10\,\mathrm{Gy}$, yielding BED = 87.5 Gy (within the published window [16]). GRACE score calibration maps to Medication intensity: Standard (GRACE < 110) / Intensified (110–140) / Maximal (> 140) [7].
- **XGBoost recommender:** 4-class (Medication-Standard / Intensified / Maximal / SBRT), n_estimators=300, max_depth=5, random_state=42.
- **Survival models:** Kaplan–Meier (non-parametric, stratified by branch, log-rank test) [19] and Cox PH (covariates: age, risk_encoded, grace_score, bed_gy; produces per-patient 2-year survival probabilities) [10].

Target: ~360 words for both parts combined.

1.4 Tan Guan Han — Results & Discussion §5

TAN GUAN HAN Results & Discussion (§f) — all sub-sections **HIGH PRIORITY**

Role. You own the full Results & Discussion section. Every number must match the briefing document exactly. Every table and figure must be referenced by number in the text. Present all results honestly — including the 1.00 XGBoost score (explain it) and the synthetic survival caveat.

Task R-1 §5.1 Part 1 Results + Comparison Table HIGH PRIORITY After models are run

What to write:

- Present **Table N**: Part 1 classifier performance on the held-out 20% test set.

Metric	Random Forest	XGBoost	Δ	Winner
Accuracy	**0.9614**	0.9579	+0.0035	RF
F1-macro	**0.9321**	0.9251	+0.0070	RF
AUC-OVR	**0.9880**	0.9853	+0.0027	RF

- State that Random Forest wins on all three metrics. Emphasise the F1-macro gap (+0.0070) as the most important difference because F1-macro is the primary selection metric [**6**].
- **Winner rationale (2 paragraphs):** (1) At ~1,400 training samples, RF's parallel bagging ensemble is more stable than XGBoost's sequential boosting — consistent with findings in [**6**] and [**13**]. (2) `class_weight='balanced'` in RF directly compensates for the 8.4% Medium class; F1-macro penalises poor recall on this minority class.
- **Figure:** Include the side-by-side confusion matrix heatmap (`reports/figures/confusion_matrix.png`) as Figure N with caption: "Confusion matrices for Random Forest (left) and XGBoost (right) on the held-out test set (285 rows, 20%). Most errors occur on the Medium (minority) class boundary, as expected."

Target: ~260 words + table + figure.

Task R-2 §5.2 Feature Importance HIGH PRIORITY After Part 1 training

What to write:

- Present the top-10 feature importances from the winning Random Forest as **Table N+1** with columns: Rank | Feature | Clinical significance. Key rows:
  - Rank 1: Oldpeak (ST depression) — direct marker of myocardial ischaemia under stress.
  - Rank 2: MaxHR — chronotropic competence.
  - Rank 3: HeartRateStressIndex (*engineered*) — validates the feature engineering step [**9**].

  - Rank 5: ChestPainType_ASY — asymptomatic ischaemia paradox.
- Write 2 sentences interpreting the HRSI result: HRSI ranking 3rd confirms that normalising MaxHR by age-predicted maximum provides information beyond either feature alone [**9**]; chronotropic competence is a recognised non-invasive ischaemia marker.
- **Figure:** Include the feature importance bar chart (`reports/figures/feature_importance.png`) as Figure N with caption: "Top-15 feature importances (mean decrease in impurity) from the winning Random Forest. The engineered HeartRateStressIndex ranks 3rd, validating the feature engineering decision."

Target: ~160 words + table + figure.

Task R-3 §5.3 Part 2 Results + Survival HIGH PRIORITY After Part 2 training

What to write:

- **Recommender results table:** XGBoost Recommender: Accuracy = 1.00, F1-macro = 1.00. **Mandatory honest note** (must appear in the report): "The perfect scores reflect the fact that intervention labels are deterministically derived from clinical routing rules (RiskCategory + arrhythmia flag + GRACE score threshold). XGBoost with max_depth=5 and 300 trees learns these exact decision boundaries. This is expected in the proof-of-concept setting; in a real MIMIC-IV deployment [**5**] labels would be drawn from noisy clinical records and scores would be substantially lower."
- **Survival results table:** Cox PH concordance C-index = 0.85 (threshold ≥ 0.65, pass). Log-rank $p < 0.05$ (SBRT vs. Medication curves differ significantly).
- **KM figure:** Include `reports/figures/km_curves.png` as Figure N with caption: "Kaplan–Meier survival curves for SBRT (red) and Medication (green) branches with 95% confidence bands. Log-rank test: $p < 0.05$, confirming significantly different survival trajectories between treatment branches [**19**]."
- **Example cases table:** Present all four cases (Cases A–D) with columns: Case | Patient | Part 1 Output | Recommendation | 2-Year Survival. Values from the briefing document Section 9.

Target: ~260 words + 2 tables + 1 figure.

Task R-4 §5.4 Discussion HIGH PRIORITY After all results written

What to write — one paragraph per point, $\geq$4 citations:

- Compare RF F1-macro = 0.9321 against Mohan et al. [**6**] (accuracy >88% on UCI alone): CardioSurv achieves competitive performance on a merged, richer, and more challenging dataset.
- Interpret the HRSI feature importance result clinically: chronotropic competence is a recognised ischaemia marker [**9**]; the HRSI captures this non-invasively from routine vital sign measurements, which is a clinically meaningful contribution of the feature engineering

step.

- Acknowledge the synthetic survival limitation transparently. Reference Wang et al. [**18**] to support the Cox PH choice over deep alternatives at this dataset scale.
- Connect to the broader clinical decision support literature [**17**]: embedding validated scoring tools (GRACE [**7**], BED [**8**]) into the ML routing layer improves clinical plausibility beyond pure data-driven models and represents a principled design choice.

Target: ~200 words, $\geq$4 citations.

2 Section 2 — Slides Assignments

#	Title	Owner	Required content
3	Datasets & Merging	Ang Jing Ru	2-row table: Kaggle (918 rows) | UCI (303 rows, adds `ca` + `thal`). Merge result: 1,221 rows. One bullet on synthetic survival rationale. Keep prose minimal — the table is the slide.
4	Feature Engineering + EDA Highlights	Ang Jing Ru	3-row table: AgeBin / BP_RiskLevel / HRSI with formulas. EDA key finding: ASY chest pain = strongest categorical predictor. 80/20 stratified split. Medium class = 8.4%. EDA class balance figure as inset if space allows.
5	Part 1 — Model Setup	Chiluba Chifunilo	Two-column layout: Random Forest hyperparameters | XGBoost hyperparameters. One line: "Both wrapped in the same sklearn Pipeline." Three metric definitions: Accuracy, F1-macro (primary), AUC-OVR. **No results on this slide.**
6	Part 1 — Results Comparison	Chiluba Chifunilo	Large comparison table (make it readable from the back of the room). RF cells highlighted in green. Delta column shown. RF wins all 3: 0.9614 / 0.9321 / 0.9880 vs 0.9579 / 0.9251 / 0.9853. 2 sentences below: why RF won (dataset scale + class weight).
7	Feature Importance + Confusion Matrix	Bougacha Mohamed	Left half: top-10 importance bar chart figure. Right half: RF confusion matrix heatmap figure. Two text annotations: "HRSI ranks #3 — validates feature engineering." and "Oldpeak #1 — direct ischaemia marker."
8	Part 2 — Recommender & Survival	Law Zi Ying	Mini-table: XGB acc=1.00 + honest 1-line note (deterministic labels). Cox C-index = 0.85. KM curves figure (inset). 4 example cases table: Cases A–D with survival before → after.

#	Title	Owner	Required content
9	Limitations	Tan Guan Han	4 bullets only. (1) Synthetic survival — proof-of-concept. (2) Small dataset (~1,400 rows). (3) No external validation cohort. (4) XGB 1.00 = deterministic labels. Tone: matter-of-fact, not apologetic.
10	Future Work	Tan Guan Han	3 bullets: (1) Real MIMIC-IV longitudinal data. (2) SHAP explainability for clinical trust. (3) Prospective pilot study vs. cardiologist assessment. Keep brief.

Note for Chiluba Chifunilo — Slide 6 (Results Comparison): This is the single most important results slide in the entire deck. Make the comparison table large enough to read from the back of the room. Highlight the Random Forest column in green. Do not just show the numbers — the 2 sentences explaining WHY RF won must be on the slide, not only said verbally.

Note for Bougacha Mohamed — Slide 7 (Feature Importance): Both figures (importance bar chart + confusion matrix) must be final print-resolution PNGs from `reports/figures/`. Do not screenshot from the notebook — use the saved PNG files. The two text annotations must be visible without zooming.

Note for Law Zi Ying — Slide 8 (Part 2 + Survival): The honest note about XGBoost acc=1.00 **must appear on the slide**, not just be said verbally. The examiner will notice it and will ask about it. Addressing it proactively demonstrates understanding rather than hiding a weakness. The suggested text: "Labels are deterministically derived from routing rules — model learns exact boundaries. Expected in proof-of-concept setting."

3 Section 3 — Reference List

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